

# Mental Math — Tricks & Drills

One-page reference for parent-led sessions • Print and keep next to the tracker

## SPEED TRICKS — the cognitive shortcuts

- BOND-TO-10 REFLEX.  $3+7$ ,  $4+6$ ,  $8+2$  must be instant retrieval, not calculation. Same for bonds to 100 ( $62+38$ ,  $73+27$ ).
- MAKE-100 SCAN. Before adding 3 numbers, scan: do any two make 100 (or 10)? Pair them first. ( $78+45+22 \rightarrow 78+22=100$ ,  $+45 = 145$ .)
- ROUND-AND-ADJUST. Anything ending in 8, 9, 11, 98, 99  $\rightarrow$  round to the friendly number, then adjust. ( $47+99 = 47+100-1 = 146$ .)
- LOOK-AHEAD CARRY. Glance at the ones BEFORE saying the tens. If they cross 10, bump tens by 1 first. Eliminates self-correction.
- COMPENSATION (SHIFTING).  $28+47 = 30+45 = 75$ . Move 2 from 47 to 28 to make a round number.
- DOUBLING. Double  $47 = 80+14 = 94$ . Once doubling is fluent, near-doubles ( $47+48$ ) become almost free.
- LTR SPOKEN CADENCE. Speak the answer as you compute: 'three hundred... eighty... six.' Cadence prevents working-memory overload.

## ACCURACY TRICKS — the discipline moves

- ESTIMATE FIRST. '478+365 is roughly 800' — say it before the exact answer. Catches wild errors instantly.
- LAST-DIGIT CHECK. The ones digit of the answer is deterministic. Verify it independently — fast and catches most errors.
- DO NOT SELF-CORRECT MID-ANSWER. Finish the answer, THEN check. Mid-correction creates a second error 30% of the time.
- REPEAT THE QUESTION before answering. Catches misheard problems, gives the brain a beat to load the numbers.
- HALF-BEAT PAUSE BEFORE THE CARRY. Most errors are at the carry, not the rest. Train a tiny pause right at that moment.
- ONE METHOD PER SESSION. In early weeks, don't switch between LTR-direct and round-and-adjust mid-problem. Flexibility comes after fluency.

## DRILLS — exercises that build both

- FLASH DRILL. One problem, 5-second timer, 20 reps. Builds retrieval speed.
- LADDER DRILL. Same starting number, climbing operand:  $47+1$ ,  $47+2$ ,  $47+3$  ...  $47+9$ . Builds carry detection.
- MAKE-100 CHAINS. 'Tell me the partner of  $73 \rightarrow \underline{\quad}$ ' (drill until under 2 sec).
- BUDDY RUNNING TOTAL. Parent calls out random 2-digit numbers; Ambar holds and updates a running sum aloud. Brutal for working memory, builds it fast.
- TWO-METHOD DRILL. Same problem solved with both LTR and round-and-adjust. Forces flexibility (use after Stages 5 & 6 are passed).
- ONE-ERROR RULE. Wrong answer = repeat the problem three times correctly before moving on. Embeds the correction.

**STOP ON A WIN.** End every session with one fast, correct problem. The last 10 seconds is what the brain consolidates overnight.